I^{2}SB through the ImMAP lens:
predict the endpoint, then resample on the bridge

ImMAP project notes

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Abstract

We give a two-line reading of the I^{2}SB sampler. The trained regressor computes the conditional expectation of the bridge's endpoint, $\hat{\boldsymbol{x}}_0 = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]$. One reverse step of I^{2}SB is then exactly: *(i)* predict the endpoint, and *(ii)* resample on the bridge between that prediction and $\boldsymbol{x}_1$ at a time $s < t$ closer to the target. The resample commits to some randomness; the next prediction conditions on it and is sharper. Along the exact reverse bridge the predictions form a martingale, from the blurry posterior mean $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$ at $t = 1$ to a single sample of $\boldsymbol{x}_0$ at $t = 0$. The only detail I^{2}SB adds is that the resample re-uses a fraction $r = \sigma(s)/\sigma(t)$ of the noise implied by the prediction; unrolled, the I^{2}SB state is a bridge sample around a precision-weighted average of all endpoint predictions so far. Resampling with entirely fresh noise ($r = 0$) is the memoryless member of the same family.

Notation. Images are vectors in $\mathbb{C}^N$; $\boldsymbol{\varepsilon}, \boldsymbol{z} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I})$. $\boldsymbol{x}_0$ is the target (the fully-sampled image), $\boldsymbol{x}_1$ the bridge's prior endpoint (in MRI: the zero-filled image $\boldsymbol{E}^H\boldsymbol{y}$, the ACS low-pass image, or a guide contrast), and $t \in [0, 1]$ the bridge time, $t = 0$ at $\boldsymbol{x}_0$. Gradients of real functions of complex vectors are Wirtinger gradients $\partial/\partial\bar{\boldsymbol{u}}$, so all formulas read as in the real case.

1 The bridge, and what the network learns

The Brownian bridge between $\boldsymbol{x}_0$ and $\boldsymbol{x}_1$ has marginals

$$\boldsymbol{x}_t = (1-t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\,\boldsymbol{\varepsilon}, \qquad \sigma_{\mathrm{sb}}(t)^2 = V\,t(1-t), \qquad \boldsymbol{\varepsilon} \perp (\boldsymbol{x}_0, \boldsymbol{x}_1), \tag{1}$$

with total variance $V = 4\tau^2$ set by the temperature τ ($\max_t \sigma_{\mathrm{sb}} = \tau$). I^{2}SB trains a regressor on pairs $(\boldsymbol{x}_0, \boldsymbol{x}_1)$, $t \sim \mathcal{U}[0, 1]$, $\boldsymbol{x}_t$ drawn from (1), with the squared loss $\mathbb{E}\|D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{x}_0\|^2$. Its minimiser is the conditional expectation,

$$D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]. \tag{2}$$

Remark 1 (General schedules). For an I^{2}SB schedule $\beta(t)$, $\boldsymbol{x}_t = \mu_0(t)\boldsymbol{x}_0 + \mu_1(t)\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\boldsymbol{\varepsilon}$ with $\mu_0 + \mu_1 = 1$. Everything below holds with $(1 - t,\, t)$ replaced by (μ_0, μ_1).

2 Step 1: predict the endpoint

At every step the network returns its prediction of the bridge's endpoint,

$$\boxed{\hat{\boldsymbol{x}}_0 \coloneqq D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1].} \tag{3}$$

This is not a small move toward $\boldsymbol{x}_0$: it is the full conditional expectation of the endpoint given everything the state reveals. How good it is depends on how much the state reveals, which the bridge time measures.

Lemma 1 (Bridge time is a noise level)**.** *Fix $\boldsymbol{x}_1$ and $t < 1$, and set $\boldsymbol{u}_t \coloneqq (\boldsymbol{x}_t - t\,\boldsymbol{x}_1)/(1-t)$. Then*

$$\boldsymbol{u}_t = \boldsymbol{x}_0 + \sigma(t)\,\boldsymbol{\varepsilon}, \qquad \sigma(t)^2 = \frac{\sigma_{\mathrm{sb}}(t)^2}{(1-t)^2} = \frac{Vt}{1-t}, \tag{4}$$

and $t \mapsto \sigma(t)$ is a bijection $[0,1) \to [0,\infty)$.

Proof. Substitute (1). □

So given $\boldsymbol{x}_1$, the state at time t is the target plus white noise of std $\sigma(t)$, and since $(\boldsymbol{x}_t, \boldsymbol{x}_1) \mapsto (\boldsymbol{u}_t, \boldsymbol{x}_1)$ is invertible, $\hat{\boldsymbol{x}}_0 = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]$ is the conditional MMSE denoiser at level $\sigma(t)$. At $t = 1$ ($\sigma = \infty$) the state carries no information beyond $\boldsymbol{x}_1$, and $\hat{\boldsymbol{x}}_0 = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$: the posterior mean over every plausible target, which is blurry. As $t \to 0$ the noise vanishes and $\hat{\boldsymbol{x}}_0 \to \boldsymbol{x}_0$.

Remark 2 (Tweedie's formula)*.* By Tweedie's formula (Miyasawa), the same prediction can be read as a gradient step on the noise-smoothed conditional prior $p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1) = \int \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{x}_1)\, d\boldsymbol{x}$:

$$\hat{\boldsymbol{x}}_0 = \boldsymbol{u}_t + \sigma(t)^2\, \nabla_{\boldsymbol{u}} \log p_{\sigma(t)}(\boldsymbol{u}_t \mid \boldsymbol{x}_1).$$

This is an identity, not an extra computation: the network computes the expectation directly, and the formula reinterprets it as a full jump of step size σ^2 up the smoothed log-density. The reading below does not need it.

3 Step 2: resample on the bridge, closer to the target

We do not stop at $\hat{\boldsymbol{x}}_0$: at large $\sigma(t)$ it is an average over many targets, not one of them. Instead we treat it as a provisional endpoint and draw a new state on the bridge between $\hat{\boldsymbol{x}}_0$ and $\boldsymbol{x}_1$, at a time $s < t$, i.e. at a smaller noise level $\sigma(s) < \sigma(t)$:

$$\boldsymbol{x}_s = (1-s)\,\hat{\boldsymbol{x}}_0 + s\,\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(s)\,\boldsymbol{\varepsilon}_s. \tag{5}$$

The noise $\boldsymbol{\varepsilon}_s$ can be drawn fresh, or partly re-used from the current state. The noise that the prediction implies is in $\boldsymbol{x}_t$ is

$$\hat{\boldsymbol{\varepsilon}}_t \coloneqq \frac{\boldsymbol{x}_t - (1-t)\,\hat{\boldsymbol{x}}_0 - t\,\boldsymbol{x}_1}{\sigma_{\mathrm{sb}}(t)} = \frac{\boldsymbol{u}_t - \hat{\boldsymbol{x}}_0}{\sigma(t)}, \tag{6}$$

and we take

$$\boldsymbol{\varepsilon}_s = r\,\hat{\boldsymbol{\varepsilon}}_t + \sqrt{1-r^2}\,\boldsymbol{z}, \qquad \boldsymbol{z} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I}) \text{ fresh}, \qquad r \in [0,1]. \tag{7}$$

We write $\boldsymbol{x}_s \sim \mathcal{R}_s(\hat{\boldsymbol{x}}_0;\, \boldsymbol{x}_t)$ for this resample, (5) with (7). If $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_0$ then $\hat{\boldsymbol{\varepsilon}}_t = \boldsymbol{\varepsilon}$ is standard normal, $\boldsymbol{\varepsilon}_s$ is too, and (5) is an exact draw from the bridge marginal at s, the input distribution the network was trained on.

The resample is where the sampler commits: the fresh noise $\boldsymbol{z}$ picks out, at the coarse scale $\sigma(s)$, one of the targets that $\hat{\boldsymbol{x}}_0$ was averaging over. The next prediction conditions on that choice.

4 The predictions sharpen: a martingale

Proposition 1 (Endpoint predictions form a martingale)**.** *Let $1 = t_0 > t_1 > \cdots > t_K = 0$, and let $\boldsymbol{x}_{t_0}, \boldsymbol{x}_{t_1}, \ldots$ be a path of the bridge (1) visited from $\boldsymbol{x}_1$ toward $\boldsymbol{x}_0$. Then*

$$\hat{\boldsymbol{x}}_0^{(k)} \coloneqq \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_{t_k}, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_{t_k}, \boldsymbol{x}_{t_{k-1}}, \ldots, \boldsymbol{x}_{t_0}], \tag{8}$$

so $(\hat{\boldsymbol{x}}_0^{(k)})_k$ is a Doob martingale: $\mathbb{E}[\hat{\boldsymbol{x}}_0^{(k+1)} \mid \boldsymbol{x}_{t_k}, \ldots, \boldsymbol{x}_{t_0}] = \hat{\boldsymbol{x}}_0^{(k)}$. It starts at the posterior mean $\hat{\boldsymbol{x}}_0^{(0)} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$ and ends at $\hat{\boldsymbol{x}}_0^{(K)} = \boldsymbol{x}_0$.

Proof. Given its endpoints, the Brownian bridge is Markov. So given $\boldsymbol{x}_{t_k}$, the states at later times $t_{k-1}, \ldots, t_0$ (closer to $\boldsymbol{x}_1$) carry no further information about $\boldsymbol{x}_0$, which gives the equality in (8). The right-hand side conditions on a growing history, so the tower property gives the martingale property. At $t_K = 0$ the state is $\boldsymbol{x}_0$. □

This is the precise sense in which each step "predicts the endpoint, and the predictions converge to it". There is no fixed $\boldsymbol{x}_0$ waiting to be found at sampling time: the sampler creates the endpoint as it goes. The first prediction is the posterior mean over all plausible targets; each resample reveals more of one particular target; each later prediction conditions on what has been revealed and is sharper; at $t = 0$ the predictions have collapsed onto a single sample of $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)$.

Remark 3 (Exact only for the true reverse bridge). Proposition 1 holds when the states are drawn from the true reverse kernel $p(\boldsymbol{x}_s \mid \boldsymbol{x}_t, \boldsymbol{x}_1) = \int q(\boldsymbol{x}_s \mid \boldsymbol{x}_t, \boldsymbol{x}_0)\, p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1)\, d\boldsymbol{x}_0$, a mixture over plausible endpoints. I^{2}SB substitutes the point prediction $\hat{\boldsymbol{x}}_0$ for $\boldsymbol{x}_0$ in $q(\boldsymbol{x}_s \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$ (Section 5). That gets the mean of $\boldsymbol{x}_s$ right but drops the spread of $\boldsymbol{x}_0$ given $\boldsymbol{x}_t$ from its variance. The gap vanishes as the steps get small.

5 I^{2}SB is this, with $r = \sigma(s)/\sigma(t)$

I^{2}SB's reverse step is the bridge posterior $q(\boldsymbol{x}_s \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$ with $\boldsymbol{x}_0$ replaced by $\hat{\boldsymbol{x}}_0$:

$$\boldsymbol{x}_s \sim \mathcal{CN}\Big((1 - \tfrac{s}{t})\hat{\boldsymbol{x}}_0 + \tfrac{s}{t}\,\boldsymbol{x}_t,\; V\,\tfrac{s(t-s)}{t}\,\boldsymbol{I}\Big) \tag{9}$$

(`sb.base.reverse_sample`, `posterior="ddpm"`).

Proposition 2 (I^{2}SB = endpoint prediction + bridge resample)**.** *The resample (5)–(7) with*

$$r = \frac{\sigma(s)}{\sigma(t)} = \sqrt{\frac{s(1-t)}{t(1-s)}} \tag{10}$$

is exactly the I^{2}SB step (9).

Proof. Substitute (6) into (5). The coefficient of $\boldsymbol{x}_t$ is $k = r\,\sigma_{\mathrm{sb}}(s)/\sigma_{\mathrm{sb}}(t)$. With $\sigma_{\mathrm{sb}} = (1-t)\sigma$ and $\sigma^2 = Vt/(1-t)$,

$$k = \frac{(1-s)\,\sigma(s)^2}{(1-t)\,\sigma(t)^2} = \frac{s}{t}.$$

The coefficient of $\boldsymbol{x}_1$ is $s - k\,t = 0$, and that of $\hat{\boldsymbol{x}}_0$ is $(1-s) - k(1-t) = (t-s)/t$: the mean of (9). The fresh-noise variance is $\sigma_{\mathrm{sb}}(s)^2(1-r^2) = Vs(1-s)\cdot\frac{t-s}{t(1-s)} = V\frac{s(t-s)}{t}$: the variance of (9). □

The identity holds for general schedules with $\sigma = \sigma_{\mathrm{sb}}/\mu_0$. We checked it numerically against `reverse_sample` for the Brownian and I^{2}SB-β schedules (agreement to 10^{-14}).

When the re-used noise matters. r depends only on how far the step moves.

- **First step from $\boldsymbol{x}_1$:** $\sigma(1) = \infty$, so $r = 0$. I^{2}SB's first step is an endpoint prediction followed by a fully fresh resample.
- **Last step to $s = 0$:** $\sigma_{\mathrm{sb}}(0) = 0$, so the output is $\hat{\boldsymbol{x}}_0$ whatever r is.
- **Coarse ladders** (few NFE): r is small (e.g. 0.27 for $t = 0.6 \to s = 0.1$), and I^{2}SB is nearly a fresh resample.
- **Fine ladders:** $r \to 1$; most of the noise is carried over and the step moves the state only slightly toward $\hat{\boldsymbol{x}}_0$.

6 What the re-used noise does: averaging the predictions

In debiased coordinates the step is especially simple. With $\boldsymbol{u}_s = \hat{\boldsymbol{x}}_0 + \sigma(s)\boldsymbol{\varepsilon}_s$ and (6)–(7),

$$\boldsymbol{u}_s = r^2\, \boldsymbol{u}_t + (1 - r^2)\, \hat{\boldsymbol{x}}_0 + \sigma(s)\sqrt{1 - r^2}\, \boldsymbol{z}. \tag{11}$$

So I^{2}SB moves the state only a fraction $1 - r^2 = 1 - \sigma(s)^2/\sigma(t)^2$ of the way to the current prediction, and re-noises to $\sigma(s)$.

Proposition 3 (Precision-weighted average)**.** *Run* (11) *down a ladder* $\sigma_0 = \infty > \sigma_1 > \cdots > \sigma_K$*, with endpoint predictions* $\hat{\boldsymbol{x}}_0^{(k)}$ *made at level* σ_k*. Then*

$$\boldsymbol{u}_K = \bar{\boldsymbol{u}}_K + \boldsymbol{n}_K, \qquad \bar{\boldsymbol{u}}_K = \sigma_K^2 \sum_{k=0}^{K-1} \Big(\frac{1}{\sigma_{k+1}^2} - \frac{1}{\sigma_k^2}\Big)\, \hat{\boldsymbol{x}}_0^{(k)}, \tag{12}$$

where the weights are nonnegative and sum to one, and $\boldsymbol{n}_K$ *is the sampler's own noise, of std* σ_K*.*

Proof. Split $\boldsymbol{u}_k = \bar{\boldsymbol{u}}_k + \boldsymbol{n}_k$ into the part driven by the $\hat{\boldsymbol{x}}_0$'s and the part driven by the injected noise. Dividing the $\bar{\boldsymbol{u}}$ part of (11) by σ_{k+1}^2 and using $r^2 = \sigma_{k+1}^2/\sigma_k^2$ gives $\bar{\boldsymbol{u}}_{k+1}/\sigma_{k+1}^2 = \bar{\boldsymbol{u}}_k/\sigma_k^2 + (\sigma_{k+1}^{-2} - \sigma_k^{-2})\, \hat{\boldsymbol{x}}_0^{(k)}$, which telescopes from $1/\sigma_0^2 = 0$. The weights sum to $\sigma_K^2(\sigma_K^{-2} - \sigma_0^{-2}) = 1$. For the noise, $\operatorname{Var} \boldsymbol{n}_{k+1} = r^4\sigma_k^2 + \sigma_{k+1}^2(1 - r^2) = \sigma_{k+1}^2$. □

This is the whole picture of I^{2}SB: *at each step, predict the endpoint; then resample on the bridge, at a lower noise level, around a running average of all endpoint predictions so far, each weighted by the precision* $1/\sigma^2$ *gained when it was made.* Early, blurry predictions get little weight; late, sharp ones get most of it. The decomposition is an algebraic identity of the sampler; the network enters only through the $\hat{\boldsymbol{x}}_0^{(k)}$, each of which depends on the state it was computed from.

Remark 4 (Fresh resampling, $r = 0$). Drawing $\boldsymbol{\varepsilon}_s$ entirely fresh gives $\boldsymbol{x}_s = (1-s)\hat{\boldsymbol{x}}_0 + s\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(s)\boldsymbol{z}$, the forward bridge sample around the latest prediction (`posterior="interpolant"`). It is memoryless: the state is centred on the last prediction only, and nothing is averaged. With a perfect network, fresh resampling and I^{2}SB both produce exact bridge marginals; they differ only in how the network's errors propagate, which makes `interpolant` vs. `ddpm` a direct test of the value of the averaging.

Algorithm 1 I^{2}SB as endpoint prediction + bridge resample

Require: $\boldsymbol{x}_1$; regressor D_θ; times $1 = t_0 > t_1 > \cdots > t_K = 0$

$\boldsymbol{x} \leftarrow \boldsymbol{x}_1$
for $k = 0, \ldots, K-1$ **do**
  $\hat{\boldsymbol{x}}_0 \leftarrow D_\theta(\boldsymbol{x}, t_k, \boldsymbol{x}_1)$ ▷ predict the endpoint
  $\boldsymbol{x} \sim \mathcal{R}_{t_{k+1}}(\hat{\boldsymbol{x}}_0; \boldsymbol{x})$ ▷ resample on the bridge
end for
return $\boldsymbol{x}$

Here $\mathcal{R}$ uses $r_k = \sigma(t_{k+1})/\sigma(t_k)$ for I^{2}SB and $r_k = 0$ for fresh resampling. The last step lands at $t_K = 0$, where $\sigma_{\text{sb}} = 0$, so the returned state is the final prediction $\hat{\boldsymbol{x}}_0$.

7 Summary

Object	Formula	Code
bridge marginal	$\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1 + \sigma_{\text{sb}}(t)\boldsymbol{\varepsilon}$, $\sigma_{\text{sb}}^2 = Vt(1-t)$	`sb.base.bridge_coeffs`
time $\leftrightarrow$ noise	$\boldsymbol{u}_t = \boldsymbol{x}_0 + \sigma(t)\boldsymbol{\varepsilon}$, $\sigma^2 = Vt/(1-t)$	—
endpoint prediction	$\hat{\boldsymbol{x}}_0 = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]$	`predict_x0`
bridge resample	$\boldsymbol{x}_s = (1-s)\hat{\boldsymbol{x}}_0 + s\boldsymbol{x}_1 + \sigma_{\text{sb}}(s)\boldsymbol{\varepsilon}_s$	`reverse_sample`
noise re-use	$\boldsymbol{\varepsilon}_s = r\hat{\boldsymbol{\varepsilon}}_t + \sqrt{1-r^2}\boldsymbol{z}$	
I^{2}SB	$r = \sigma(s)/\sigma(t)$	`posterior="ddpm"`
fresh	$r = 0$	`posterior="interpolant"`
predictions	martingale, $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] \to \boldsymbol{x}_0$ (exact reverse bridge)	—
I^{2}SB state	precision-weighted average of the $\hat{\boldsymbol{x}}_0^{(k)}$, plus noise	—

Table 1: I^{2}SB as endpoint prediction followed by a bridge resample (`sb/base.py`).

References

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